Submission 3

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Data Science Final Project Submission 3

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Preeclampsia, Vitamin D, and Pregnancy Duration

Submission notes:

Resources & References:

Source article: https://pmc.ncbi.nlm.nih.gov/articles/PMC5127689/

RASAL3 Gene: https://www.genecards.org/card/RASAL3

packageCheck function by Carly Bobak

https://danmaclean.github.io/tidyversebook/03-joining.html

ggplot2 cheat sheet: https://www.maths.usyd.edu.au/u/UG/SM/STAT3022/r/current/Misc/data-visualization-2.1.pdf

Pre-defined themes for *ggplot2*: https://ggplot2.tidyverse.org/reference/ggtheme.html

https://r-graph-gallery.com/263-ggplot2-boxplot-parameters.html

https://stackoverflow.com/questions/22036227/list-as-input-of-a-function

Data notes:

Initial sample size: Final sample size: Analyses conducted:

Setup

Packages

```
knitr::opts_chunk$set(echo = TRUE)

# Packages to use
# tidyverse includes ggplot2
# janitor and DataExplorer are good for data wrangling
packages = c("tidyverse", "janitor", "DataExplorer", "ggtext", "corrplot",
```

```
             "ggridges", "pheatmap", "table1", "RColorBrewer")

packageCheck = function(x) {
    if (!require(x, character.only = TRUE)) {
      install.packages(x, dependencies = TRUE)
      library(x, character.only = TRUE)
    }
  }

# Now load or install & load all
loadPacks<-lapply(packages, packageCheck)
```

```
## Loading required package: tidyverse
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
## Loading required package: janitor
##
##
## Attaching package: 'janitor'
##
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
##
##
## Loading required package: DataExplorer
##
## Loading required package: ggtext
##
## Loading required package: corrplot
##
## corrplot 0.95 loaded
##
## Loading required package: ggridges
##
## Loading required package: pheatmap
##
## Loading required package: table1
##
##
## Attaching package: 'table1'
##
##
```

```
## The following objects are masked from 'package:base':
##
##     units, units<-
##
##
## Loading required package: RColorBrewer
```

```
remotes::install_github("hrbrmstr/waffle")
```

```
## Using GitHub PAT from the git credential store.
## Skipping install of 'waffle' from a github remote, the SHA1 (767875bf) has not changed since last ins
##   Use `force = TRUE` to force installation
```

```
library("waffle")
```

```
## Registered S3 method overwritten by 'plyr':
##   method    from
##   [.indexed table1
```

```
set.seed(10)
```

Versions

ggplot2: 4.0.3

tidyverse: 2.0.0

janitor: 2.2.1

DataExplorer: 0.9.0

ggtext: 0.2.0

corrplot

ggridges

waffle ?

pheatmap:

table1:

Data Setup

Load in data and save to raw data files

```
# Covariate data
rawCovar <- read.csv(file="GSE85307_covar-1.csv", sep = ",")
# Gene expression data
rawGeneExp <- read.csv(file="GSE85307_geneExpression.csv", sep = ",")
# Gene names
rawGeneNames <- read.csv(file="GSE85307_geneNames.csv", sep = ",")
```

Confirm Data Appears as Expected and Examine Setup

```
#introduce(rawCovar)
#head(rawCovar)

#introduce(rawGeneExp)
#head(rawGeneExp)

#introduce(rawGeneNames)
#head(rawGeneNames)
```

Save to new data frames

Save raw data to new data frames to be able to reset to raw data without re-importing original files. View new data frames.

```
covar <- rawCovar
geneExp <- rawGeneExp
geneNames <- rawGeneNames

#view(covar)
#view(geneExp)
#view(geneNames)
```

Data Wrangling

Clean covariate data

Covariate data includes metadata in values spaces that needs to be removed. Some column names can be shortened for clarity and concision.

```
cleanCovar <- covar %>%
  separate(col="mat_age_yrs", into = c(NA, "mat_age"), sep = "\\ ",
           convert = TRUE) %>%
  separate(col="preg_preeclampsia", into = c(NA, "preg_preeclampsia"),
           sep = "\\ ", convert = TRUE) %>%
  separate(col="delivery_gestAge_wks", into = c(NA, "delivery_weeks"),
           sep = "\\ ", convert = TRUE) %>%
  separate(col="sampleCollection_gestAge_wks", into = c(NA, "sample_weeks"),
           sep = "\\ ", convert = TRUE) %>%
  separate(col="batch_Microarray", into = c(NA, "batch_Microarray"),
           sep = "\\ ", convert = TRUE) %>%
  separate(col="mat_BMI", into = c(NA, "mat_BMI"), sep = "\\ ",
           convert = TRUE) %>%
  separate(col="mat_vitD", into = c(NA, "mat_vitD"), sep = "\\ ",
           convert = TRUE) %>%
  separate(col="mat_asthma", into = c(NA, "mat_asthma"), sep = "\\ ",
           convert = TRUE)
```

Add gene names to gene expression data

```
named_genes <- geneExp %>%
  rename(PROBEID = X) %>%
  left_join(geneNames, by = "PROBEID")
rownames(named_genes) <- named_genes$GENE
named_genes <- named_genes[,-1] # remove PROBEID column
#named_genes <- named_genes[,-158] # remove GENE column
```

Pivot named genes into a long data frame

```
long_genes <- named_genes %>%
  pivot_longer(col = c(starts_with("GSM")), values_to = "geneExp",
               names_to = "geoID")
```

Join covariate data and gene expression data

```
long_cov_genes <- long_genes %>%
  left_join(cleanCovar, by = "geoID")
```

Turn two categorical covariates into factors

```
# Preeclampsia status
long_cov_genes$preg_preeclampsia <- factor(long_cov_genes$preg_preeclampsia)

# Maternal asthma status
long_cov_genes$mat_asthma <- factor(ifelse(long_cov_genes$mat_asthma == 1,
                                           "Asthma", "Not Asthma"))
```

Turn one continuous covariate into a categorical covariate

Remember to include this information in labels

```
# Delivery weeks in buckets
# Very Early: before 36 weeks
# Early: between 36 and 38 weeks
# On time: between 38 and 40 weeks
# Late: after 40 weeks

long_cov_genes$delivery_time <- factor(
  ifelse(long_cov_genes$delivery_weeks < 36, "Very Early",
         ifelse(long_cov_genes$delivery_weeks < 38, "Early",
                ifelse(long_cov_genes$delivery_weeks > 40, "Late", "On Time"))),
  levels = c("Very Early", "Early", "On Time", "Late"), ordered = TRUE)
levels(long_cov_genes$delivery_time)
```

```
## [1] "Very Early" "Early"      "On Time"    "Late"
```

```
summary(long_cov_genes$delivery_time)
```

```
## Very Early      Early    On Time       Late
##       4000      12000      43500      19000
```

Do this for cleanCovar as well

```
cleanCovar$delivery_time <- factor(
  ifelse(cleanCovar$delivery_weeks < 36, "VeryEarly",
         ifelse(cleanCovar$delivery_weeks < 38, "Early",
                ifelse(cleanCovar$delivery_weeks > 40, "Late", "OnTime"))),
  levels = c("VeryEarly", "Early", "OnTime", "Late"), ordered = TRUE)
```

Histogram Function

```
# Histogram
create_histogram <- function(df, geneName){
  # x is gene expression
  ggplot(df, aes(x = geneExp)) +
    geom_histogram(bins = 40) +
    labs(
      title = paste0(geneName," Gene Expression in Subset of VDAART Study"),
      x = paste0("Gene Expression of ", geneName),
      y = "Frequency" # find units
    ) +
    theme_bw() +
    theme(
      plot.title = element_textbox_simple(
        padding = margin(t = 10, b = 10)
      )
    )
}
```

Scatterplot Function

```
create_scatterplot <- function(df, geneName, contCov){
  # x is continuous covariate, y is gene expression
  # covariate currently hard coded
  ggplot(df, aes(x = mat_vitD, y = geneExp)) +
    geom_point() +
    labs(
      title = paste0("Maternal Vitamin D vs. Gene Expression of ", geneName,
                     " at Trial Entry (10-18 Weeks Gestation)"),
      x = "Maternal Vitamin D (ng/mL)", # contCov
      y = paste0("Expression of ", geneName) # geneName
    ) +
    geom_smooth(method = "loess", formula = y ~ x) +
```

```
    theme_bw() +
    theme(
      plot.title = element_textbox_simple(
        padding = margin(t = 10, b = 10)
      )
    )
}
```

Boxplot Function

With jitter

```
create_boxplot <- function(df, geneName, catCov1, catCov2){
  # y is gene expression, x is delivery time in buckets,
  # color is preeclampsia status
  # preeclampsia and delivery time currently hard coded
  boxplot <- ggplot(df, aes(x = delivery_time, y = geneExp,
                            color = preg_preeclampsia)) +
    geom_boxplot(outlier.shape = NA) +
    labs(
      y = geneName,
      x = "Pregnancy Duration in Weeks",
      color = "Preeclampsia Status",
      title = paste0("Gene Expression of ", geneName,
                     " with Pregnancy Duration and Preeclampsia Status")
    ) +
    theme_bw() +
    scale_x_discrete(
      name = "Pregnancy Duration in Weeks",
      labels = c("< 36", "36-38", "38-40", "> 40")
    ) +
    scale_color_discrete(
      name = "Preeclampsia Status",
      labels = c("Normal", "Preeclampsia")
    ) +
    geom_point(position = position_jitterdodge()) +
    #geom_jitter(width = 0.2) +
    theme(
      plot.title = element_textbox_simple(
        padding = margin(t = 10, b = 10)
      ),
      axis.text.y = element_text(face="italic")
    )

  return(boxplot)
}
```

Define function to create all three plots for a given gene

```
create_plot <- function(df, geneName, contCov, catCov1, catCov2){
  # get only relevant parts of the dataframe
```

```
  gene <- df[which(df["GENE"] == geneName),]
  # save plots to vector to be returned as one object
  plots <- c()

  # still passing geneName separately so it's easier to get for titles and labels

  plots <- append(plots, create_histogram(gene, geneName))
  plots <- append(plots, create_scatterplot(gene, geneName, contCov))
  plots <- append(plots, create_boxplot(gene, geneName, catCov1, catCov2))
  return(plots)
}
```

Define function to iterate over list of genes

```
# function to create all of three plots for each gene name given
create_plots <- function(df, geneNames, contCov = "mat_vitD",
                         catCov1 = "delivery_time",
                         catCov2 = "preg_preeclampsia"){
  # save plots to vector to be returned as one object
  emptyPlots <- c()

  # iterate over list of genes
  for (i in geneNames) {
    # create all three types of plots for each gene
    emptyPlots <- append(emptyPlots, create_plot(df, i, contCov, catCov1, catCov2))
  }

  return(emptyPlots)
}
```

Call function to produce plots

Genes chosen: RASAL3 plays a role in signalling and T cell regulation. It is associated with diverticulitis and epiphyseal dysplasia.

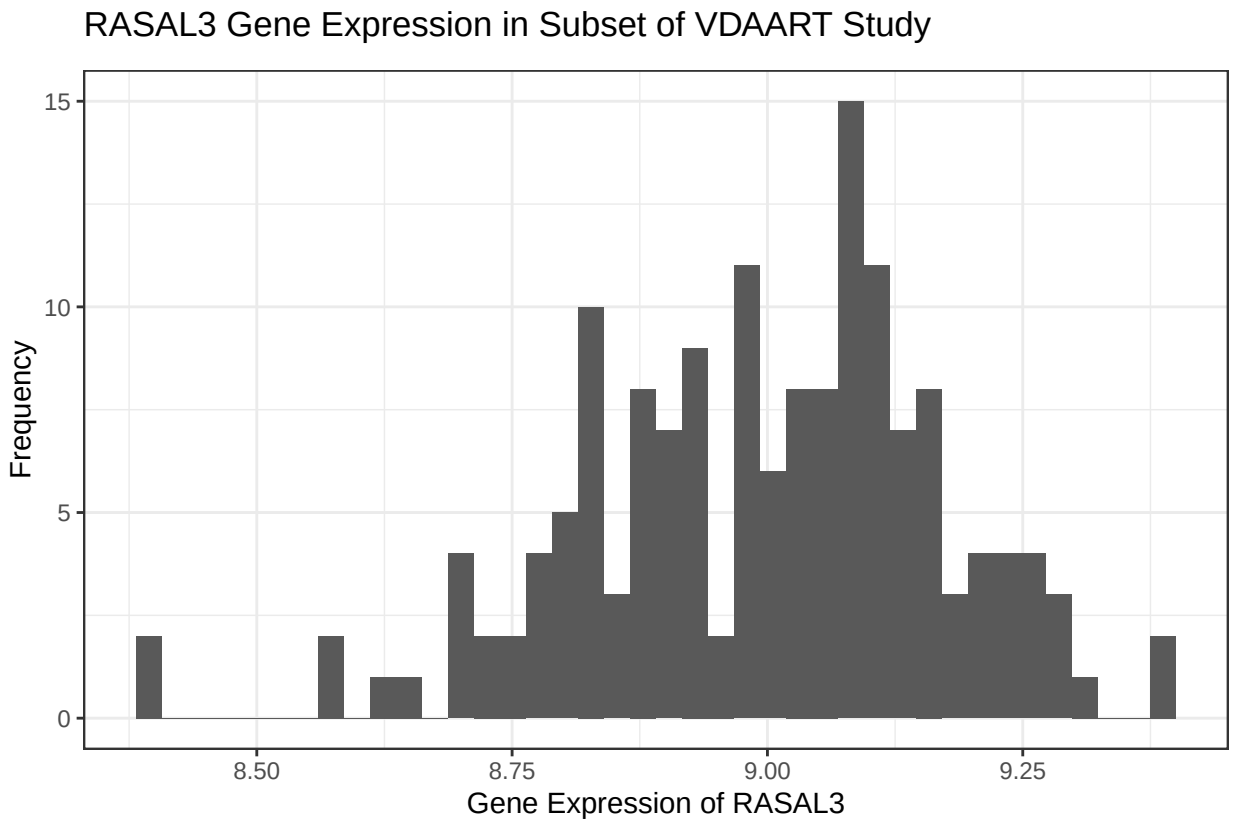
```
# List of genes to examine
geneNames = c("RASAL3", "UNC13A","UNC13C", "IL2", "IL24", "IL26", "IL7",
              "RBM15", "RBM47", "RBM7")
# so. several of these actually do not have useful information :/ time to pick different ones
# was hoping for groups of similar genes
# original list: "RASAL3", "UNC13A","UNC13C", "IL12RB1", "IL13", "IL1F10",
#"IL2", "IL24", "IL26", "IL7"
```

For Submission 3, this is only being run for RASAL3 in order to not create 30 plots.

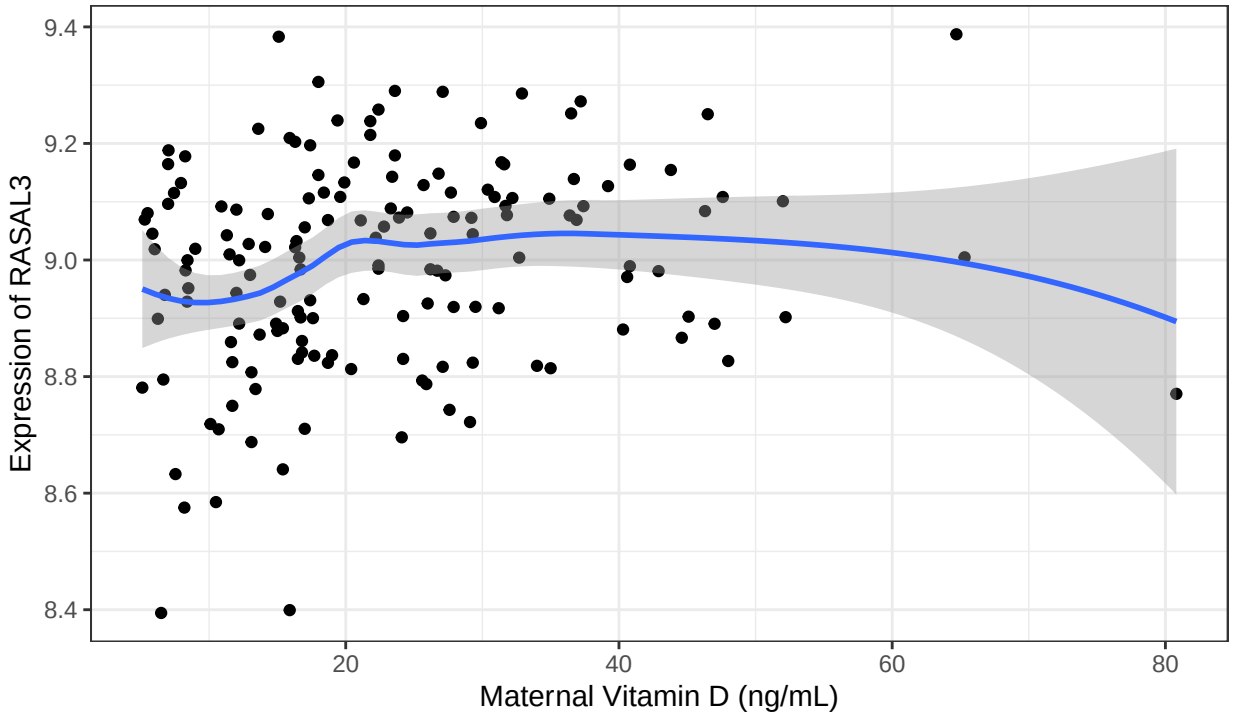
```
# Call function to make plots
allPlots <- create_plots(df = long_cov_genes, geneNames = geneNames[1],
                         contCov = "mat_vitD", catCov1 = "delivery_time",
                         catCov2 = "preg_preeclampsia")
```

```
# Show plots
allPlots
```

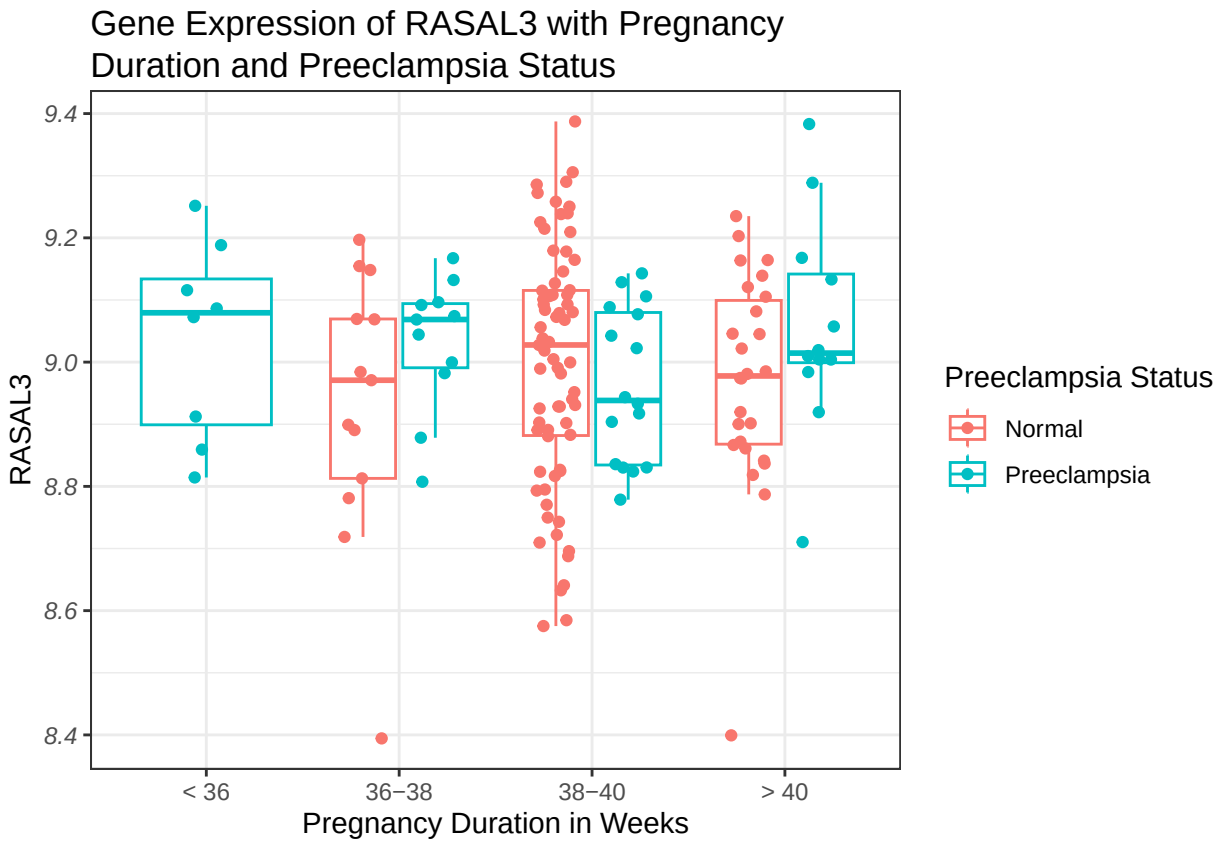
```
## [[1]]
```




```
##
## [[2]]
```


Maternal Vitamin D vs. Gene Expression of RASAL3 at Trial Entry (10−18 Weeks Gestation)
Expression of RASAL3
9.4
9.2
9.0
8.8
8.6
8.4
20
40
60
80
Maternal Vitamin D (ng/mL)

```
##
## [[3]]
```



Subset by selected genes

```
chosen_genes <- subset(
  named_genes, (GENE == geneNames[1] | GENE == geneNames[2] | GENE == geneNames[3]
                | GENE == geneNames[4] | GENE == geneNames[5] | GENE == geneNames[6]
                | GENE == geneNames[7] | GENE == geneNames[8]
                | GENE == geneNames[9] | GENE == geneNames[10]
                )
  )

chosen_long_genes <- as.data.frame(
  subset(long_cov_genes, (GENE == geneNames[1] | GENE == geneNames[2]
                          | GENE == geneNames[3] | GENE == geneNames[4]
                          | GENE == geneNames[5] | GENE == geneNames[6]
                          | GENE == geneNames[7] | GENE == geneNames[8]
                          | GENE == geneNames[9] | GENE == geneNames[10])
         )
  )

chosen_cov_genes <- subset(
  long_cov_genes, (GENE == geneNames[1] | GENE == geneNames[2] | GENE == geneNames[3]
                   | GENE == geneNames[4] | GENE == geneNames[5] | GENE == geneNames[6]
                   | GENE == geneNames[7] | GENE == geneNames[8] | GENE == geneNames[9]
                   | GENE == geneNames[10]
                   )
```

```
  )

rownames(cleanCovar) <- cleanCovar$geoID
cleanCovar <- cleanCovar[,-2]

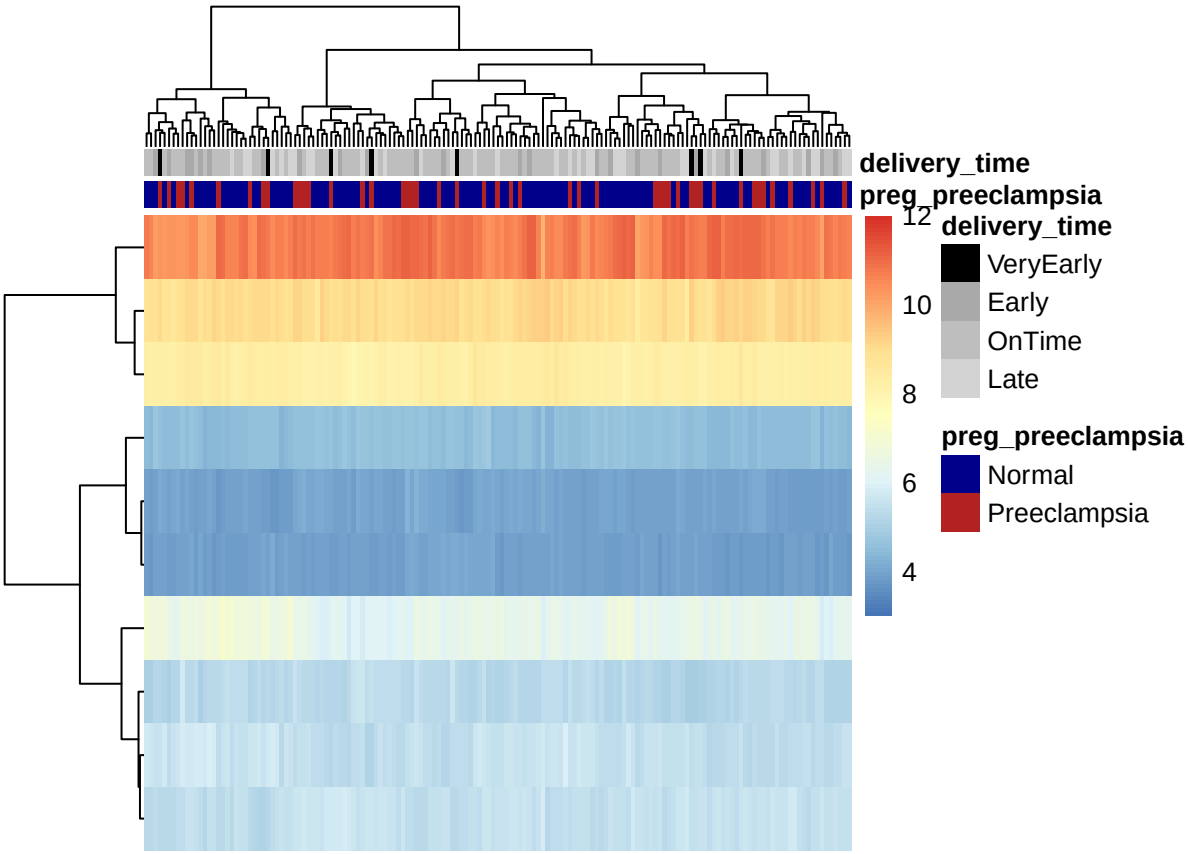
# remove column with gene names which leaves all numeric columns
chosen_genes <- chosen_genes[,-158]
```

Heatmap

```
column_annot <- as.data.frame(cleanCovar[,c("preg_preeclampsia", "delivery_time")])
rownames(column_annot) <- rownames(cleanCovar)

pheat_colors <- list(
  `preg_preeclampsia` = c(Normal="darkblue",Preeclampsia="firebrick"),
  `delivery_time` = c(VeryEarly= "black", Early= "darkgray",
                      OnTime = "gray", Late = "lightgray")
)

pheatmap(chosen_genes,
         show_rownames = F,
         show_colnames = F,
         annotation_col = column_annot,
         annotation_colors = pheat_colors,
         clustering_distance_cols = "canberra",
         clustering_distance_rows = "canberra",
         clustering_method = "ward.D2",
         breaks=seq(from=3, to=12, length.out=101))
```



Additional Plot: Ridge Plot

```
ggplot(chosen_cov_genes, aes(x = geneExp, y = GENE, fill = GENE)) +
  geom_density_ridges() +
  labs(
    title = "Distributions of Expressions of Selected Genes",
    x = "Gene Expression",
    y = "Gene"
  ) +
  theme_ridges() +
  theme(
    legend.position = "none",
    axis.text.y = element_text(face="italic"),
    axis.title.y = element_text(hjust = 0.5),
    axis.title.x = element_text(hjust = 0.5)
    )
```

```
## Picking joint bandwidth of 0.0494
```

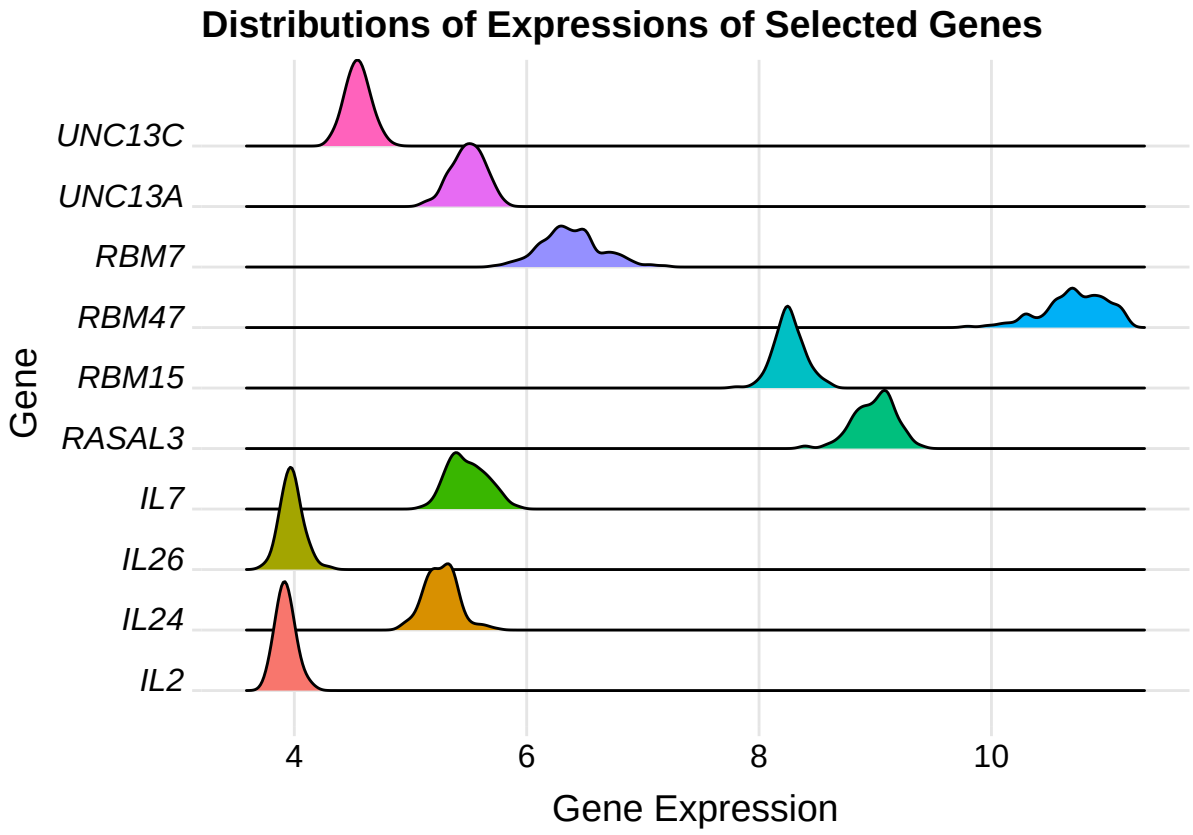


Table 1

```
# Labels were modified for LaTeX
table1(~ mat_age + mat_vitD + mat_BMI + delivery_time |
         preg_preeclampsia, long_cov_genes)
```

	Normal (N=55000)	Preeclampsia (N=23500)	Overall (N=78500)
mat_age			
Mean (SD)	26.6 (5.06)	25.8 (4.90)	26.4 (5.03)
Median [Min, Max]	26.9 [18.5, 38.8]	25.7 [18.4, 35.3]	26.6 [18.4, 38.8]
mat_vitD			
Mean (SD)	24.4 (14.5)	19.7 (8.24)	23.0 (13.1)
Median [Min, Max]	21.8 [5.11, 80.8]	17.7 [7.00, 36.5]	20.4 [5.11, 80.8]
mat_BMI			
Mean (SD)	28.0 (7.63)	31.2 (7.94)	28.9 (7.86)
Median [Min, Max]	26.5 [18.0, 61.9]	31.2 [18.3, 54.4]	27.3 [18.0, 61.9]
delivery_time			
Very Early	0 (0%)	4000 (17.0%)	4000 (5.1%)
Early	6500 (11.8%)	5500 (23.4%)	12000 (15.3%)
On Time	35500 (64.5%)	8000 (34.0%)	43500 (55.4%)
Late	13000 (23.6%)	6000 (25.5%)	19000 (24.2%)